search_api.py ×

```python
import requests

def search(search_term='luke'):
  base_url = 'https://swapi.dev/api/people/?search='
  search_url = f'{base_url}{search_term}'
  resp = requests.get(search_url)
  resp_json = resp.json()
  if resp_json.get('results'):
    return resp.json()['results'][0]
  else:
    return None

if __name__ == '__main__':
  import pprint

  character = search()
  pprint.pprint(character)
```

12% (2:1)

Terminal ×

```
7.22)
(sw) codio@cabinetchef-igloofocus:~/workspace$ python sw_characters/search_api.py
{'birth_year': '19BBY',
 'created': '2014-12-09T13:50:51.644000Z',
 'edited': '2014-12-20T21:17:56.891000Z',
 'eye_color': 'blue',
 'films': ['https://swapi.dev/api/films/1/',
           'https://swapi.dev/api/films/2/',
           'https://swapi.dev/api/films/3/',
           'https://swapi.dev/api/films/6/'],
 'gender': 'male',
 'hair_color': 'blond',
 'height': '172',
 'homeworld': 'https://swapi.dev/api/planets/1/',
 'mass': '77',
 'name': 'Luke Skywalker',
 'skin_color': 'fair',
 'species': [],
 'starships': ['https://swapi.dev/api/starships/12/',
               'https://swapi.dev/api/starships/22/'],
 'url': 'https://swapi.dev/api/people/1/',
 'vehicles': ['https://swapi.dev/api/vehicles/14/',
              'https://swapi.dev/api/vehicles/30/']}
(sw) codio@cabinetchef-igloofocus:~/workspace$
```

Guide ×

Collapse

Searching the API

Initial Search

Activate the virtual environment.

```
conda activate sw
```

While the *Star Wars* API is very broad, we are going to focus on just the characters from the films. You can access a specific character by referencing a specific URL and the site returns the relevant information in JSON. For example, the page below is for Luke Skywalker:

```
https://swapi.dev/api/people/1/
```

The problem is, you need to know the exact URL for each character if want to access their information. The API also provides a way to search it. This too is done with a specific URL. Replace the number with `?search=r2` where `r2` represents the search term. The API will still return information as JSON.

```
https://swapi.dev/api/people/?search=r2
```

Start by importing `requests`. Create the `search_sw` function with `luke` as the default value for `search_term`. For now, put `pass` as the body of the function. In addition, we are going to set up a space to test our code. Use a conditional to ask if we are running the script directly. Use `pass` as the body of the conditional for now.

```python
import requests

def search(search_term='luke'):
  pass

if __name__ == '__main__':
  pass
```

Every search starts with the same base URL. Create the variable `base_url` that contains the search URL minus the search term. Then create the `search_url` variable that combines the `base_url` and the `search_term` argument.

```python
def search(search_term='luke'):
  base_url = 'https://swapi.dev/api/people/?search='
  search_url = f'{base_url}{search_term}'
```

The `resp` variable represents the response from searching the API. Use the `get()` method from the `requests` package and the `search_url` as the website to request. Then transform the response into JSON with the `json()` method. Python treats JSON like a dictionary, which means we can easily access the keys and values.

```python
def search(search_term='luke'):
  base_url = 'https://swapi.dev/api/people/?search='
  search_url = f'{base_url}{search_term}'
  resp = requests.get(search_url)
  resp_json = resp.json()
```

If the API call worked and returned information, it will be stored in the `results` key. The associated value is a list of dictionaries. Use a conditional to determine if the value for `results` is an empty list or not. An empty list means the character is not found in the API. If the list is not empty, return the first element from `results`, which is a dictionary. If `results` is an empty list, return `None`.

```python
def search(search_term='luke'):
  base_url = 'https://swapi.dev/api/people/?search='
  search_url = f'{base_url}{search_term}'
  resp = requests.get(search_url)
  resp_json = resp.json()
  if resp_json.get('results'):
    return resp.json()['results'][0]
  else:
    return None
```

Testing the Function

Let's test out the newly created function. In the testing conditional import the `pprint` package. This allows us to print the contents of a dictionary in an easy to read manner. Create `character` and set it to the return value from `search_sw`. By default, the function will search for `'luke'`. Use `pprint` to print `character`.

```python
if __name__ == '__main__':
  import pprint

  character = search()
  pprint.pprint(character)
```

In the terminal, run the script to test your work.

```
python sw_characters/search_api.py
```

You should see the following output:

```
{'birth_year': '19BBY',
 'created': '2014-12-09T13:50:51.644000Z',
 'edited': '2014-12-20T21:17:56.891000Z',
 'eye_color': 'blue',
 'films': ['https://swapi.dev/api/films/1/',
           'https://swapi.dev/api/films/2/',
           'https://swapi.dev/api/films/3/',
           'https://swapi.dev/api/films/6/'],
 'gender': 'male',
 'hair_color': 'blond',
 'height': '172',
 'homeworld': 'https://swapi.dev/api/planets/1/',
 'mass': '77',
 'name': 'Luke Skywalker',
 'skin_color': 'fair',
 'species': [],
 'starships': ['https://swapi.dev/api/starships/12/',
               'https://swapi.dev/api/starships/22/'],
 'url': 'https://swapi.dev/api/people/1/',
 'vehicles': ['https://swapi.dev/api/vehicles/14/',
              'https://swapi.dev/api/vehicles/30/']}
```

Now, test the function with a search term that does not exist, `'asdf'`.

```python
if __name__ == '__main__':
  import pprint

  character = search('asdf')
  pprint.pprint(character)
```

In the terminal, run the script to test your work.

```
python sw_characters/search_api.py
```

You should see the following output:

```
None
```

▼ **Code**

Your code should look like this:

```python
import requests

def search(search_term='luke'):
  base_url = 'https://swapi.dev/api/people/?search='
  search_url = f'{base_url}{search_term}'
  resp = requests.get(search_url)
  resp_json = resp.json()
  if resp_json.get('results'):
    return resp.json()['results'][0]
  else:
    return None

if __name__ == '__main__':
  import pprint

  character = search('asdf')
  pprint.pprint(character)
```

Deactivate the virtual environment.

```
conda deactivate
```

Lab Question

Select the commands to build a wheel for the package called `my_package` with both Conda and Poetry. **Hint**, there is more than one correct answer.